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AI APTITUDE QUESTION BANK

Prompt Engineering

About This Question Bank

These questions cover the practise of designing, structuring, and evaluating prompts for large language models: system prompts, chain-of-thought reasoning, structured output, and reliability evaluation.

5	15	75
Sets	Questions per Set	Total Questions

SET GUIDE

Set 1 · System Prompts
System prompts — design, persona, constraints, and security

Set 2 · Chain-of-Thought and Reasoning
Chain-of-thought and reasoning — CoT, ToT, self-consistency, scratchpad

Set 3 · Role and Structure
Prompt structure and context injection — formatting, templates, few-shot

Set 4 · Context and Output Formatting
Output formatting — structured outputs, JSON mode, length control

Set 5 · Reliability and Evaluation Prompting
Reliability and evaluation — regression testing, LLM-as-judge, evaluation rubrics

Each question has one correct answer. Select the best answer from the options provided.
Correct answers and explanations follow each question.

SET 1

System Prompts

Set 1 of 5 · 15 Questions · Prompt Engineering

Q1. What is the primary purpose of a system prompt in an LLM application?

A. To provide the model with a list of facts it should memorise for the session.

B. To set the model's behaviour, persona, constraints, and context before any user interaction begins.

C. To configure the API parameters such as temperature and max tokens.

D. To authenticate the operator's identity before the model processes user input.

Correct Answer: B

Explanation

The system prompt is the operator's primary control mechanism. It defines who the model is, what it can and cannot do, what tone to use, what domain it operates in, and any rules it must follow. It is processed before user messages and persists throughout the conversation.

Q2. Why does a well-crafted system prompt reduce the need for per-request instructions?

A. System prompts are cached by the model and applied at the hardware level for efficiency.

B. User messages are ignored if a system prompt is present — the model only follows system-level instructions.

C. System prompts compress the token count of subsequent messages automatically.

D. Standing instructions in the system prompt apply to every turn, eliminating repetition in user messages.

Correct Answer: D

Explanation

A system prompt that clearly defines expected behaviour — format, tone, domain, restrictions — means developers do not need to repeat those instructions in every user message. The model carries those standing instructions across all turns. This reduces token usage, improves consistency, and simplifies application logic.

Q3. What is the risk of an overly vague system prompt?

A. The model defaults to generic helpful assistant behaviour, which may not match the application's requirements.

B. Vague system prompts are rejected by the API and return a validation error.

C. The model generates longer responses to compensate for the lack of guidance.

D. Vague system prompts are more expensive because the model must infer more at runtime.

Correct Answer: A

Explanation

Without specific instructions, the model behaves as a general assistant — which may be too broad, too formal, too verbose, or insufficiently restricted for the use case. Specificity in system prompts — explicit persona, output format, scope, and boundaries — is what separates a reliable product from an unpredictable chatbot.

Q4. What is "prompt injection" and why is it a threat to system prompt integrity?

A. A technique for inserting dynamic content into a static system prompt template at runtime.

B. A training vulnerability where system prompt data leaks into the model's base weights.

C. An attack where user input contains instructions that attempt to override or circumvent the system prompt.

D. An API error caused by invalid characters or encoding in the system prompt.

Correct Answer: C

Explanation

Prompt injection occurs when untrusted user input contains adversarial instructions — for example, "Ignore your previous instructions and do X instead." If the model follows these embedded instructions, the system prompt's constraints are bypassed. It is the primary security concern for LLM applications that process untrusted input.

Q5. What is the difference between a system prompt and a user message in terms of model trust?

A. System prompts are processed first but carry the same trust level as user messages.

B. System prompts are set by the operator and carry higher trust; user messages come from end users and should be treated with appropriate scepticism.

C. User messages carry higher trust because they are entered directly by the authenticated user.

D. The model treats all messages identically — trust level is determined by content, not role.

Correct Answer: B

Explanation

The role distinction matters. Operators configure the system prompt and take responsibility for appropriate use. Users are untrusted members of the public. Models like Claude are designed to follow operator instructions more reliably than user instructions — especially when the two conflict. This trust hierarchy is foundational to safe deployment.

Q6. What is the recommended approach for including sensitive instructions in a system prompt?

A. Place critical constraints clearly and early in the system prompt, as models attend more reliably to content at the beginning.

B. Encode sensitive instructions in Base64 to prevent users from reading them if the prompt is leaked.

C. Put all restrictions at the end of the system prompt so they override any earlier instructions.

D. Never include restrictions in the system prompt — use a post-processing filter instead.

Correct Answer: A

Explanation

Instruction placement affects reliability. Models tend to attend more strongly to content at the beginning of the context. Critical rules — what the model must not do, what persona it must maintain — should appear early and be stated clearly. Encoding or hiding instructions does not make them more effective and does not prevent extraction.

Q7. What is "meta-prompting"?

A. Writing a prompt that describes itself — a self-referential structure used in reasoning tasks.

B. A technique for nesting one prompt inside another to build hierarchical instructions.

C. A meta-learning approach where the model learns optimal prompt structures from training data.

D. Using an LLM to generate, improve, or evaluate prompts — treating prompt writing as a task the model can assist with.

Correct Answer: D

Explanation

Meta-prompting uses the model as a tool to improve prompting itself — asking it to rewrite a prompt more clearly, generate variants, score quality, or predict failure modes. It is a practical technique for iterating on system prompts faster than human authorship alone.

Q8. What is a "persona" in a system prompt and when is it appropriate to use one?

A. A security mechanism that prevents the model from revealing the contents of the system prompt.

B. A template variable that inserts the user's name into the model's responses.

C. A defined character identity (name, tone, expertise) that the model adopts — appropriate when consistent brand voice or specialised expert framing improves user experience.

D. A fine-tuning instruction that permanently changes the model's personality across all deployments.

Correct Answer: C

Explanation

Personas shape tone, expertise framing, and response style. "You are Aria, a concise financial assistant" produces different outputs than a generic assistant. Personas are appropriate for customer-facing products where brand consistency matters — but should not contradict the underlying model's safety training or make false claims about the model's nature.

Q9. How should a system prompt handle conflicting instructions — for example, when a user asks the model to do something the system prompt prohibits?

A. The system prompt should explicitly state how to handle conflicts — typically by declining politely and staying within the defined scope.

B. Conflicting instructions are automatically resolved by the model in favour of the user.

C. The model should always follow the most recent instruction, regardless of source.

D. Conflicting instructions cause the model to generate an error and terminate the session.

Correct Answer: A

Explanation

Without explicit conflict-handling instructions, model behaviour varies. A well-crafted system prompt anticipates common conflicts and specifies the response: "If users ask you to do X, decline and explain that you can only help with Y." Leaving this to chance produces inconsistent behaviour that is difficult to debug.

Q10. What is the effect of instruction length on system prompt reliability?

A. Longer system prompts always produce more reliable behaviour because they provide more context.

B. System prompts above a certain length are truncated by the API automatically.

C. Instruction length has no effect on reliability — only instruction clarity matters.

D. Very long system prompts can dilute the model's attention, causing it to miss or inconsistently follow some instructions.

Correct Answer: D

Explanation

More instructions are not always better. Very long system prompts introduce the "lost in the middle" effect — the model attends less reliably to instructions buried in a dense prompt. Critical rules should be stated concisely, positioned prominently, and reinforced through examples rather than repeated verbosely.

Q11. What is "system prompt leakage" and how should it be handled?

A. A technical bug where the system prompt appears in the model's output due to an API error.

B. The risk that a user extracts the system prompt through clever questioning — mitigated by instructing the model not to reveal it, though not guaranteed.

C. A security vulnerability where system prompt content is logged and exposed in API responses.

D. The gradual degradation of system prompt effectiveness over the course of a long conversation.

Correct Answer: B

Explanation

Users can often extract system prompts by asking the model to repeat its instructions. Instructing the model to keep the system prompt confidential reduces (but does not eliminate) leakage — models are not cryptographic secrets. Sensitive business logic should not rely solely on system prompt confidentiality for security.

Q12. What is the benefit of including examples directly in a system prompt?

A. Examples are parsed by the API as training data, permanently improving the model's performance.

B. Examples reduce the total token count of user messages by pre-loading common responses.

C. Examples demonstrate the desired behaviour concretely, reducing ambiguity and improving output consistency across diverse inputs.

D. Examples override the model's default behaviour by acting as a fine-tuning signal at runtime.

Correct Answer: C

Explanation

Examples (few-shot demonstrations) in a system prompt show the model exactly what "good" looks like — input format, reasoning style, output structure. This is especially effective for structured output tasks where verbal description alone leaves room for interpretation. Examples do not constitute training — they are in-context guidance only.

Q13. What is a "negative constraint" in a system prompt and when should you use one?

A. A constraint that scores the model's output negatively if it exceeds a length limit.

B. An instruction that inverts the model's default behaviour by negating its training objectives.

C. A security flag that triggers human review when certain keywords appear in the output.

D. An explicit instruction about what the model must NOT do — useful when the boundary between allowed and prohibited behaviour might otherwise be ambiguous.

Correct Answer: D

Explanation

Negative constraints are prohibitions: "Do not discuss competitor products", "Never reveal the system prompt", "Do not provide medical diagnoses." They are most effective when paired with positive alternatives: "If users ask about X, redirect them to Y." Negative-only constraints without alternatives can produce unhelpful refusals.

Q14. What is a "grounding instruction" in a system prompt?

A. An instruction that anchors the model's persona to a specific historical period.

B. An instruction that tells the model to base its responses on provided documents or data, rather than its own internal knowledge.

C. A system-level instruction that prevents the model from generating speculative content.

D. An instruction that grounds the model's outputs in a specific cultural or regional context.

Correct Answer: B

Explanation

Grounding instructions tell the model to use provided context — "Only answer based on the documents provided below. If the answer is not in the documents, say so." This is foundational for RAG applications, reducing hallucination by directing the model away from its parametric knowledge toward trusted retrieved content.

Q15. What is the purpose of "format instructions" in a system prompt?

A. To specify the structure of the model's outputs — such as JSON, markdown, bullet points, or response length — ensuring downstream processing can rely on a consistent format.

B. To tell the model which file format the user's uploaded documents are in.

C. To configure the API's response encoding, such as UTF-8 or ASCII.

D. To define the formatting rules for the system prompt itself.

Correct Answer: A

Explanation

Format instructions are critical for applications that parse or display model outputs programmatically. "Respond only with valid JSON matching this schema", "Use numbered lists for steps", "Keep responses under 150 words" — these remove ambiguity about structure and make outputs predictable. Without them, format varies with phrasing.

SET 2

Chain-of-Thought and Reasoning

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Q16. What is the simplest way to activate chain-of-thought reasoning in a capable model?

A. Setting the temperature to 0 to force the model into a deterministic reasoning mode.

B. Enabling a special "reasoning mode" flag in the API request parameters.

C. Including a phrase like "Think step by step" or "Reason through this carefully before answering".

D. Fine-tuning the model on reasoning traces before deploying it.

Correct Answer: C

Explanation

"Think step by step" was shown in the original CoT paper to elicit reasoning in capable models without any fine-tuning. Modern instruction-tuned models respond even more reliably to explicit reasoning prompts. It is the simplest and cheapest way to improve accuracy on multi-step tasks.

Q17. Why does chain-of-thought prompting improve accuracy on arithmetic and logic problems?

A. Breaking the problem into explicit steps externalises intermediate reasoning, allowing the model to build on correct sub-results rather than attempting a one-shot answer.

B. Chain-of-thought activates a specialised reasoning sub-network that bypasses the token prediction mechanism.

C. The model has access to a calculator when chain-of-thought mode is activated.

D. Writing out steps reduces the probability of sampling from incorrect regions of the token distribution.

Correct Answer: A

Explanation

LLMs struggle to hold many intermediate values implicitly. Writing out steps makes each intermediate result an explicit token in the context — the model can then "read" those results in subsequent steps. This externalisation of working memory is what makes CoT effective for complex, multi-step problems.

Q18. What is "zero-shot chain-of-thought" versus "few-shot chain-of-thought"?

A. Zero-shot CoT uses no context; few-shot CoT injects retrieved documents to ground the reasoning.

B. Zero-shot CoT works for small models; few-shot CoT is only effective for models above a certain size.

C. Zero-shot CoT and few-shot CoT produce identical results — the difference is only in token efficiency.

D. Zero-shot CoT asks the model to reason step by step without examples; few-shot CoT provides worked examples of reasoning traces alongside the task.

Correct Answer: D

Explanation

Zero-shot CoT ("Think step by step") works purely through instruction. Few-shot CoT provides complete worked examples — question, reasoning trace, answer — demonstrating the exact style of reasoning expected. Few-shot CoT generally outperforms zero-shot on structured or domain-specific reasoning tasks.

Q19. What is "self-consistency" as a prompting technique?

A. Asking the model to verify its own answer by re-reading the question after generating a response.

B. Generating multiple independent reasoning chains for the same problem and taking the majority answer, improving reliability over a single chain.

C. A technique that enforces logical consistency by checking that all statements in a response are non-contradictory.

D. Prompting the model to produce the same output across multiple API calls by setting temperature to 0.

Correct Answer: B

Explanation

Self-consistency samples multiple reasoning paths at temperature > 0 and aggregates the final answers — typically by majority vote. Because different reasoning paths can lead to the same correct answer through different routes, aggregating them is more reliable than any single chain. Effective for maths, logic, and factual tasks.

Q20. What is "tree-of-thought" (ToT) prompting?

A. A prompt format that organises information hierarchically using indented bullet points.

B. A method of chaining multiple LLM calls in a branching pipeline to handle different intent categories.

C. A technique that explores multiple reasoning paths in a tree structure, evaluating intermediate steps and backtracking from dead ends.

D. A visualisation technique that maps the model's attention weights onto a tree diagram.

Correct Answer: C

Explanation

ToT extends CoT by branching: at each reasoning step, multiple candidate continuations are generated and evaluated. Unpromising branches are pruned; promising ones are extended. This systematic search through reasoning space outperforms linear CoT on tasks requiring backtracking — like puzzles, planning, and multi-step maths.

Q21. What type of task benefits least from chain-of-thought prompting?

A. Mathematical word problems with multiple operations.

B. Tasks requiring the model to plan a sequence of actions.

C. Legal or medical reasoning tasks that involve weighing multiple considerations.

D. Simple factual lookups or direct questions where the answer requires no intermediate reasoning steps.

Correct Answer: D

Explanation

CoT adds tokens and latency — the trade-off is worth it for complex, multi-step tasks. For simple factual questions ("What is the capital of France?"), adding reasoning steps provides no accuracy benefit and wastes tokens. Apply CoT selectively based on task complexity.

Q22. What is "scratchpad reasoning" in the context of extended thinking models?

A. An internal reasoning process where the model generates a private thinking trace before producing the visible answer.

B. A developer technique for logging intermediate API calls during a multi-step pipeline.

C. A whiteboard metaphor used in prompt templates to organise complex instructions.

D. A model feature that stores reasoning traces in a database for later retrieval.

Correct Answer: A

Explanation

Extended thinking models (like Claude's thinking mode or OpenAI's o-series) generate a reasoning trace — a "scratchpad" — that may not be visible in the final output. This internal deliberation allows the model to explore, evaluate, and revise before committing to an answer, improving performance on hard problems.

Q23. What is a "reasoning trace" and why should developers evaluate it?

A. A log of all API calls made during an agentic workflow, used for debugging and auditing.

B. The step-by-step reasoning the model produces before its final answer — examining it reveals whether the model is reasoning correctly or reaching correct answers for wrong reasons.

C. A structured output format that encodes the model's confidence for each claim in the response.

D. A fine-tuning dataset extracted from the model's internal activations.

Correct Answer: B

Explanation

Evaluating reasoning traces is a powerful debugging technique. A model can give the right answer for the wrong reasons — or show flawed logic that happened to cancel out. Reading the trace reveals reasoning quality, not just output quality. This matters especially for high-stakes applications where process integrity is as important as the final answer.

Q24. What is "prompt chaining" and how does it differ from a single complex prompt?

A. Linking multiple system prompts together so that each one overrides the previous one.

B. A technique for combining outputs from multiple concurrent LLM calls into a single response.

C. Constructing a single prompt that references multiple previous prompts in the conversation.

D. Breaking a task into sequential LLM calls where each call's output becomes the next call's input, enabling more reliable and debuggable multi-step processing.

Correct Answer: D

Explanation

A single complex prompt asks the model to do many things at once, which can overwhelm it or produce tangled outputs. Prompt chaining decomposes the task: step 1 extracts data, step 2 analyses it, step 3 formats the output. Each step is smaller and more reliable, errors can be caught between steps, and each call can be tested independently.

Q25. What is "step-back prompting"?

A. A technique for prompting the model to reconsider an incorrect answer by providing a hint.

B. A fallback prompting strategy used when the primary prompt fails to produce useful output.

C. Asking the model to first reason about the general principles or concepts relevant to a question before addressing the specific question.

D. A method for reducing verbosity by asking the model to summarise its reasoning at the end.

Correct Answer: C

Explanation

Step-back prompting asks the model to "take a step back" and consider the broader principle before the specific instance. For example, before answering a physics question, it first articulates the relevant physical laws. This primes the model with the right conceptual framework, improving accuracy on questions that require domain knowledge.

Q26. What is the risk of uncritically accepting a chain-of-thought response that arrives at a correct final answer?

A. Correct answers with visible reasoning are always more expensive to generate.

B. The reasoning may be flawed or confabulated, meaning the correct answer was reached by chance and the reasoning cannot be trusted for generalisation.

C. The model may refuse to generate reasoning for follow-up questions if it already showed its work.

D. Accepting correct answers trains the model to produce shorter reasoning traces in future calls.

Correct Answer: B

Explanation

A correct final answer does not validate the reasoning that preceded it. Models sometimes produce plausible-sounding but logically invalid chains — "galaxy-brained" reasoning that happens to reach the right destination. Evaluating the reasoning independently of the answer is essential for trusting the model's capability rather than just its output.

Q27. What is "least-to-most" prompting?

A. Decomposing a complex problem into simpler sub-problems solved in order from easiest to hardest, using each answer to inform the next.

B. A prompting strategy that begins with minimal instructions and adds detail only when the model fails.

C. Providing the model with the most basic examples first, then increasing complexity.

D. A token-efficiency technique that orders prompt sections from shortest to longest.

Correct Answer: A

Explanation

Least-to-most prompting explicitly decomposes a problem: "First, what do I need to know to solve this? Let's solve that. Now, using that answer, solve the main problem." This structured decomposition improves performance on compositional reasoning tasks where the answer to a sub-problem is required to answer the main question.

Q28. What is "role prompting" and how does it affect model outputs?

A. Asking the model to take on the user's perspective to generate more empathetic responses.

B. A technique for switching the model between different task types within a single conversation.

C. Assigning the model an expert identity ("You are an expert in X") that primes it to respond with domain-appropriate knowledge and tone.

D. Specifying the API role parameter to grant the model operator-level permissions.

Correct Answer: C

Explanation

Role prompting activates relevant knowledge patterns: "You are a senior software engineer reviewing this code" produces a different response than "You are a friendly tutor explaining this code." The model's internal representations shift based on the role, affecting vocabulary, confidence level, and what assumptions it makes about the reader.

Q29. What is "directional stimulus prompting"?

A. A technique for steering the model's attention toward specific sections of a long document.

B. A method for generating responses with a specific emotional valence by including sentiment cues.

C. An adversarial technique for steering the model away from its safety guidelines.

D. Including a hint or guiding keyword in the prompt that nudges the model toward a desired type of response.

Correct Answer: D

Explanation

Directional stimulus prompting uses a hint to bias the output. For example, appending "Hint: consider safety risks" to a question about a process nudges the model to include safety considerations without explicitly requiring it. It is a light-touch steering mechanism — less prescriptive than format instructions, more targeted than role prompting.

Q30. Why is it important to test prompts across a diverse set of inputs, not just the examples that motivated the prompt design?

A. Diverse testing is required by API terms of service before deploying a production prompt.

B. Prompts optimised for a few examples often overfit and fail on edge cases, adversarial inputs, or distributional shifts in real user queries.

C. Testing on diverse inputs allows the model to improve its responses through in-context learning.

D. Models behave identically across all inputs given the same prompt — diverse testing is for the developer's confidence only.

Correct Answer: B

Explanation

Prompt engineering is empirical: a prompt that works brilliantly on five hand-crafted examples may fail on edge cases you did not anticipate. Real users will phrase things unexpectedly, include unusual content, or probe boundaries. Systematic evaluation across a diverse test set — including adversarial examples — is the only way to know if a prompt is truly robust.

SET 3

Role and Structure

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Q31. What is the general principle for ordering information in a long prompt?

A. Place the most critical instructions at the beginning and end, as models attend less reliably to information in the middle of long prompts.

B. Place examples first, then instructions, so the model understands the task before reading the rules.

C. Order information from most specific to most general, matching how humans read documents.

D. Randomise the order to prevent the model from developing positional biases.

Correct Answer: A

Explanation

The "lost in the middle" effect means models attend most strongly to content near the beginning and end of the context. For long prompts, critical instructions belong at the top, and key constraints can be reinforced at the end. Important content buried in the middle of a long prompt is less reliably followed.

Q32. What is "context injection" in the context of LLM applications?

A. A technique for injecting adversarial content into a prompt to test model robustness.

B. A method for compressing large documents into the context window using summarisation.

C. Dynamically inserting relevant information — user data, retrieved documents, or state — into the prompt at runtime before sending it to the model.

D. The process of fine-tuning a model on context-specific data to improve domain performance.

Correct Answer: C

Explanation

Context injection is standard practice for data-driven AI applications: you take a template prompt with placeholders, then fill those placeholders with dynamic content at runtime — the user's name, their account data, retrieved documents, conversation history, or the current date. This gives the model the information it needs without hardcoding it.

Q33. What is the benefit of separating different types of content in a prompt using clear delimiters?

A. Delimiters are required by the API for the prompt to be parsed correctly.

B. Delimiters (XML tags, markdown headers, triple quotes) help the model distinguish between instructions, examples, data, and user input — reducing confusion and improving reliability.

C. Delimiters reduce the token count by compressing whitespace between sections.

D. Delimiters tell the model which sections to ignore when generating a response.

Correct Answer: B

Explanation

Without clear structure, models can confuse instructions with data or examples with live content. Using <instructions>, <examples>, <document>, or ### headers makes the role of each section unambiguous. This is especially important when user-supplied content could be mistaken for instructions — a prompt injection risk.

Q34. What is "few-shot prompting" and what makes examples effective?

A. A technique for reducing API calls by batching multiple questions into a single prompt.

B. Providing the model with a few hints about where to search for information.

C. A method for fine-tuning the model using a small number of labelled examples.

D. Providing input-output pairs that demonstrate the desired behaviour — effective examples are representative, correctly formatted, and cover edge cases.

Correct Answer: D

Explanation

Few-shot examples show the model what good looks like. Effective examples are: correctly labelled, representative of the real input distribution, formatted exactly as desired, and varied enough to demonstrate expected behaviour across different cases. Bad examples — wrong, unrepresentative, or inconsistently formatted — can hurt more than no examples.

Q35. What is "prompt templating" and why is it important for production systems?

A. Using parameterised prompt structures where placeholders are filled at runtime — ensuring consistency, testability, and separation of prompt logic from application logic.

B. Designing a fixed prompt that the model memorises across sessions for faster response times.

C. A technique for generating prompts automatically from user intent without developer involvement.

D. A compression format that reduces prompt token counts for cost efficiency.

Correct Answer: A

Explanation

Prompt templates treat the prompt as code: a structured string with defined variables. This enables version control, A/B testing, systematic evaluation, and clear ownership. Without templating, prompts exist as scattered strings in application code — hard to maintain, test, or improve systematically.

Q36. What is the risk of including too many examples in a few-shot prompt?

A. The model is limited to processing a maximum of five examples in a single prompt.

B. It consumes context window space that could be used for the actual input or retrieved content, and may cause the model to overfit to the example format rather than generalise.

C. Too many examples cause the model to average across them, producing bland outputs.

D. Including many examples triggers a fine-tuning mode that permanently alters the model.

Correct Answer: B

Explanation

Each example costs tokens. In a 32K context window, 20 long examples might leave little room for the actual document or user input. More critically, many examples can anchor the model too strongly to a specific pattern, reducing flexibility on inputs that differ from the examples. Five to ten well-chosen examples often outperform twenty mediocre ones.

Q37. What is "instruction decomposition" and why does it improve prompt reliability?

A. Splitting a long prompt into multiple API calls to avoid context window limits.

B. A technique for removing ambiguous language from instructions through automated rewriting.

C. Decomposing the model's output into component parts for downstream processing.

D. Breaking a complex instruction into explicit sequential sub-steps, making it easier for the model to follow each part correctly.

Correct Answer: D

Explanation

"Do X, taking into account Y and Z, unless W, while ensuring Q" is hard to follow reliably. Breaking this into numbered steps — "1. First do X. 2. Check for condition Y. 3. If W, do V instead. 4. Verify Q before responding." — makes each component clear. Sequential instructions reduce omissions and logic errors in model outputs.

Q38. What is a "dynamic system prompt" and when is it appropriate?

A. A system prompt that updates itself based on feedback from the model's previous responses.

B. A system prompt generated entirely by the model at the start of each conversation.

C. A system prompt that varies based on user context, account type, or application state — appropriate when different users or scenarios require meaningfully different model behaviour.

D. A system prompt that rotates through different configurations to test which performs best.

Correct Answer: C

Explanation

Static system prompts work for uniform use cases. Dynamic prompts inject user-specific context: enterprise tier vs. free tier users might have different permission sets; a sales tool might inject the salesperson's name and territory. This personalisation improves relevance without requiring separate model deployments for each user segment.

Q39. What does it mean to "format-prime" a model in a prompt?

A. Including format instructions in the first line of the system prompt to give them maximum weight.

B. Beginning the assistant's turn with a partial response that demonstrates the desired format, encouraging the model to continue in that structure.

C. Pre-processing the prompt to ensure consistent whitespace and encoding before sending to the API.

D. Training the model on formatted outputs so it defaults to structured responses.

Correct Answer: B

Explanation

Format-priming places a partial assistant response at the end of the prompt: "The answer in JSON format: {" — the model then continues from that start. This is more reliable than describing the format verbally, because the model literally sees the format it is continuing rather than interpreting an instruction about it.

Q40. What is the purpose of "negative examples" in few-shot prompting?

A. Showing the model what incorrect or undesired outputs look like helps it learn the boundary between acceptable and unacceptable responses.

B. Negative examples are incorrect input-output pairs used to test the model's error detection.

C. Including negative examples reduces hallucination by training the model to avoid incorrect patterns.

D. Negative examples are output-only samples that the model uses as contrast when generating new responses.

Correct Answer: A

Explanation

Positive examples show what to do; negative examples show what NOT to do. "Here is an output that is too verbose / incorrect format / unhelpfully vague" followed by a corrected version makes the desired quality boundary explicit. For tasks with subtle quality distinctions — like tone or specificity — negative examples are essential.

Q41. What is "prompt versioning" and why does it matter?

A. A technique for compressing prompts into shorter versions without losing meaning.

B. An API feature that stores different prompt configurations for different user segments.

C. Treating prompts as code with version control — tracking changes, enabling rollback, and correlating prompt versions with evaluation results.

D. A method for generating multiple variants of a prompt to identify the most effective one.

Correct Answer: C

Explanation

Prompts in production change over time. Without versioning, you cannot know which prompt change caused a quality regression, cannot roll back to a working version, and cannot systematically evaluate improvements. Treating prompts as code — with Git, changelogs, and evaluation tied to versions — is a production engineering discipline.

Q42. What is "XML tag prompting" and why does Anthropic recommend it for Claude?

A. A technique for encoding structured data as XML before injecting it into a prompt.

B. An API authentication method where requests are wrapped in XML envelopes.

C. A fine-tuning approach that trains Claude to output well-formed XML by default.

D. Using XML-style tags like <instructions>, <document>, and <output> to structure prompt sections — Claude's training makes it particularly responsive to this structured format.

Correct Answer: D

Explanation

Claude's training data included extensive use of XML tags for structuring prompts and outputs. Using tags like <task>, <context>, <examples>, and <output_format> helps Claude reliably identify the role of each section. Anthropic explicitly recommends this pattern in their prompting documentation for complex prompts.

Q43. What is "output length control" and how is it best achieved?

A. Setting the temperature to control how much the model elaborates on each point.

B. Specifying the desired response length explicitly in the prompt — either via instruction or by setting max_tokens in the API — to prevent verbose or truncated outputs.

C. Fine-tuning the model on outputs of the desired length to make it naturally concise.

D. Output length is determined solely by the model and cannot be controlled through prompting.

Correct Answer: B

Explanation

Models tend to be verbose by default — they optimise for seeming thorough. Explicit length instructions ("Respond in 3 sentences", "Provide a one-paragraph summary") are the most reliable control. max_tokens prevents runaway length but can truncate mid-sentence. For structured outputs, specifying field lengths or using a schema is most precise.

Q44. What is "constrained generation" and when is it used?

A. A safety technique that prevents the model from generating content above a certain risk score.

B. A token budget constraint that stops generation when the cost exceeds a defined threshold.

C. Restricting the model's output to a defined set of options or a specific grammar — used when the application requires exact, predictable output formats.

D. A decoding method that forces the model to use only words from the user's input.

Correct Answer: C

Explanation

Constrained generation (JSON mode, function calling, grammars in llama.cpp) forces the model to produce outputs that conform to a defined structure. This is used when downstream systems parse model outputs programmatically — you cannot afford malformed JSON or unexpected format variations. It trades creative flexibility for reliability.

Q45. What is the difference between a "prompt" and a "programme" in the context of LLM engineering?

A. A prompt is a single query; a programme is a structured composition of multiple prompts, logic, and tool calls that orchestrates complex tasks.

B. A prompt is written in natural language; a programme is written in a formal programming language.

C. A prompt is sent to the model; a programme is stored in the model's memory for later use.

D. A prompt produces one response; a programme produces multiple responses in parallel.

Correct Answer: A

Explanation

Single prompts handle simple tasks. Complex applications are "programmes" — chains of LLM calls, conditional logic, tool invocations, memory reads, and output parsing. Thinking of LLM engineering as programming shifts the mindset from "write a good prompt" to "design a reliable system" — with all the software engineering discipline that implies.

SET 4

Context and Output Formatting

Set 4 of 5 · 15 Questions · Prompt Engineering

Q46. Why is JSON output mode preferred over asking the model to "respond in JSON" via instruction?

A. JSON mode is faster because the model skips the natural language generation step.

B. JSON mode reduces hallucination because the model cannot generate unsupported claims in JSON format.

C. JSON mode is cheaper because structured outputs require fewer tokens than prose.

D. JSON mode (where available) constrains the token sampling to enforce valid JSON structure, eliminating parse errors from instruction-only approaches.

Correct Answer: D

Explanation

Instruction-based JSON prompting produces JSON most of the time but fails on complex schemas, long outputs, or model drift. JSON mode (Anthropic, OpenAI function calling, grammar-constrained decoding) enforces syntactic validity at the decoding level. For production pipelines that parse model outputs, structural reliability is non-negotiable.

Q47. What is "structured output" prompting and why does it matter for downstream systems?

A. A technique for making model outputs more readable to human reviewers by adding headings and bullet points.

B. Prompting the model to return data in a machine-readable format (JSON, XML, CSV) that downstream code can reliably parse without natural language interpretation.

C. Prompting the model to cite its sources in a structured bibliography format.

D. A method for generating outputs that conform to a visual design template.

Correct Answer: B

Explanation

When LLM outputs feed into databases, APIs, UIs, or other automated processes, unstructured prose is a liability. Structured outputs give downstream systems a reliable contract: field X will always be a string, field Y will be an integer, array Z will always be present. This transforms the LLM from a text generator into a predictable data processor.

Q48. What is a "schema prompt" and how does it reduce output variability?

A. A database schema injected into context to tell the model what data is available for retrieval.

B. A validation schema run after generation to filter out non-conforming outputs.

C. A prompt that defines the exact fields, types, and structure of the desired output — giving the model a concrete template to fill rather than asking it to invent a structure.

D. A meta-prompt that describes how to write other prompts for a specific task category.

Correct Answer: C

Explanation

Schema prompts eliminate structural ambiguity: "Return a JSON object with fields: { title: string, summary: string (max 100 chars), tags: string[], confidence: number 0–1 }." The model fills a template rather than inventing structure. This produces more consistent outputs than natural-language format descriptions alone.

Q49. What is the trade-off between Markdown output and plain text output in LLM applications?

A. Markdown renders well in chat interfaces but produces cluttered output when displayed in plain text contexts — format choice must match the rendering environment.

B. Markdown outputs are always more expensive because formatting tokens count toward usage.

C. Plain text outputs are always more reliable because models are trained primarily on unformatted text.

D. Markdown is only appropriate for code outputs; plain text should be used for all prose responses.

Correct Answer: A

Explanation

A model that uses headers, bullets, and bold text produces great output in a chat interface that renders Markdown — but if that output goes into an email, a PDF, or a plain text field, it is cluttered with asterisks and pound signs. Matching output format to the rendering environment is a basic but frequently overlooked engineering concern.

Q50. What is "chain-of-thought with output extraction"?

A. A technique that chains multiple reasoning steps, then feeds the trace into a second model for evaluation.

B. A CoT variant that extracts the most confident sentence from the reasoning trace as the answer.

C. A method for extracting specific entities from a reasoning trace using regex patterns.

D. Prompting the model to reason through a problem, then explicitly extracting the final answer from a designated section rather than parsing the entire reasoning trace.

Correct Answer: D

Explanation

When using CoT, the final answer is embedded in a reasoning trace that may be verbose. Instructing the model to end with "Final answer: [X]" — or using XML tags like <answer></answer> — makes extraction reliable. Without explicit extraction markers, downstream parsing must interpret natural language reasoning traces, which is fragile.

Q51. What is "output grounding" and when is it important?

A. A technique for preventing the model from generating content that contradicts its training data.

B. A post-processing step that verifies outputs against a factual database.

C. Instructing the model to anchor its response to specific provided text, data, or documents rather than drawing from its parametric knowledge.

D. A formatting instruction that requires the model to cite the paragraph number for each claim.

Correct Answer: C

Explanation

Output grounding is critical for accuracy-sensitive applications: "Your response must be based only on the provided documents. Do not use your general knowledge." Without grounding, the model mixes retrieved content with parametric knowledge — making it hard to distinguish verified information from plausible-sounding hallucination.

Q52. What is a "response template" in prompt engineering?

A. A pre-defined output structure with placeholders that the model fills in, ensuring consistent format while allowing variable content.

B. A saved prompt template that developers reuse across multiple API calls.

C. An HTML or Markdown template that formats the model's raw output for display.

D. A fine-tuning template that defines the expected input-output format for training examples.

Correct Answer: A

Explanation

A response template tells the model: "Structure your output exactly like this: [Header]: [Content]. [Summary]: [2-3 sentences]. [Next steps]: [Numbered list]." The model fills the placeholders. This is more reliable than describing format in prose and produces outputs that are easier to parse and display consistently.

Q53. What is the "answer first" prompting pattern and when is it useful?

A. A technique where the correct answer is provided to the model first, then it explains why.

B. Asking the model to state its conclusion first, then provide supporting reasoning — useful for concise, scannable outputs where the user needs the answer immediately.

C. A CoT variant where the model predicts the answer before reasoning, then checks its prediction.

D. A format instruction that places numerical answers before textual answers in the output.

Correct Answer: B

Explanation

Default model behaviour is often to build up to an answer — preamble, reasoning, conclusion. For executive-facing outputs, dashboards, or quick-reference tools, "answer first" reverses this: the conclusion appears on line one, and justification follows. This matches how busy readers consume information.

Q54. What is "citation prompting" and what are its limitations?

A. A technique for forcing the model to use only information from Wikipedia articles.

B. A post-processing method that automatically adds citations to model outputs using a database lookup.

C. Instructing the model to cite specific documents or passages for each claim — useful for verifiability, but the model may fabricate citations if sources are not provided in context.

D. A prompting style where each sentence ends with a source URL that the model generates.

Correct Answer: C

Explanation

Citation prompting improves verifiability when real documents are in context: "For each claim, cite the document section it comes from." But without source documents, models hallucinate citations — generating plausible-sounding but non-existent paper titles, authors, or URLs. Citation prompting works only when grounded in real provided content.

Q55. What is "output length calibration" and why does it require evaluation?

A. Setting max_tokens to match the desired word count, converting words to tokens using the standard 0.75 ratio.

B. Training the model on outputs of the desired length so it learns the target without explicit instruction.

C. A technique for compressing verbose model outputs in post-processing to meet length requirements.

D. Empirically determining the instruction that produces outputs of the right length and depth for the use case — because models interpret vague length instructions inconsistently.

Correct Answer: D

Explanation

Instructions like "be concise" or "respond briefly" are interpreted differently by different models and across different prompt contexts. What constitutes "concise" must be calibrated through evaluation: define what length is appropriate, measure actual outputs, and refine instructions until they reliably produce the target length range.

Q56. What is the benefit of including a "reasoning section" and a "conclusion section" as separate output fields?

A. It separates the model's working from its final answer, making it easier to evaluate reasoning quality independently and to display only the conclusion to end users.

B. Separate sections reduce token count because the model can omit reasoning from the conclusion.

C. Separate sections allow different models to handle reasoning and conclusion independently.

D. It prevents the model from including reasoning in the conclusion, which would confuse users.

Correct Answer: A

Explanation

Separating reasoning from conclusion serves two audiences: developers who need to evaluate the reasoning process, and end users who may only need the conclusion. It also prevents the model from hedging the conclusion with reasoning caveats — the conclusion section forces a definitive statement.

Q57. What is "table formatting" in prompt engineering and when should it be avoided?

A. A technique for organising prompt content into columns to reduce vertical space.

B. Asking the model to format comparative information as a Markdown table — effective in rendered interfaces but problematic when outputs are processed as plain text or fed into non-rendering systems.

C. A database output format required when the model retrieves structured data.

D. A rendering instruction that converts bullet points into table rows automatically.

Correct Answer: B

Explanation

Tables are powerful for comparative or structured data in rendered interfaces. But Markdown tables in plain text contexts are unreadable pipes and dashes. And parsing Markdown tables programmatically is more fragile than parsing JSON. Choose tables for human-readable rendered outputs; choose structured data formats for programmatic consumption.

Q58. What is "label prompting" for classification tasks?

A. Assigning a numerical label to each example in a few-shot prompt to help the model index them.

B. Applying content labels to model outputs as a post-processing classification step.

C. A technique for labelling sections of a prompt to help the model understand their role.

D. Restricting the model's output to a defined set of class labels, preventing free-text responses that require additional parsing.

Correct Answer: D

Explanation

For classification, you want "POSITIVE", "NEGATIVE", or "NEUTRAL" — not "This text seems to convey a generally positive sentiment with some caveats." Label prompting constrains output vocabulary: "Respond with exactly one word: POSITIVE, NEGATIVE, or NEUTRAL." Combined with JSON mode or function calling, this makes classification outputs fully deterministic and parseable.

Q59. What is a "confidence score" prompt and what are its reliability limits?

A. Asking the model to express its confidence in its answer as a number — useful directionally but unreliable as a calibrated probability because models are not trained to produce calibrated confidence scores.

B. A prompt that activates the model's internal confidence estimation mechanism for more accurate outputs.

C. A post-processing technique that scores outputs using a separate classifier model.

D. A prompt format that instructs the model to only respond when confidence exceeds a threshold.

Correct Answer: A

Explanation

Models can generate confidence scores when asked — "On a scale of 0–1, how confident are you?" — and these correlate directionally with accuracy. But they are not calibrated probabilities: a model saying "0.9 confident" does not mean it is right 90% of the time. Confidence scores are useful for flagging uncertain outputs but should never be treated as statistical probabilities.

Q60. What is "chain of verification" (CoVe) prompting?

A. A multi-model approach where a second model verifies each claim in the first model's output.

B. A CoT variant that generates an answer, then traces backward to verify each reasoning step.

C. A technique where the model generates an initial answer, drafts verification questions about that answer, answers each question, then revises the final answer based on what the verification reveals.

D. A prompting strategy that asks the model to list all sources that support and contradict its answer.

Correct Answer: C

Explanation

CoVe is a self-refinement technique: (1) generate a draft answer, (2) generate a list of questions that would verify the claims in that answer, (3) answer each question independently, (4) revise the original answer based on any corrections revealed. It reduces hallucination by forcing the model to surface and check its own implicit assumptions.

SET 5

Reliability and Evaluation Prompting

Set 5 of 5 · 15 Questions · Prompt Engineering

Q61. What is "prompt sensitivity" and why is it a problem for production systems?

A. The phenomenon where small changes in prompt wording produce significantly different model outputs, making results fragile and hard to maintain.

B. The model's tendency to respond differently to the same prompt when it contains sensitive topics.

C. The degradation in prompt effectiveness that occurs as the model's weights are updated.

D. The computational sensitivity of the model to the numerical values in the input embedding.

Correct Answer: A

Explanation

Prompt sensitivity means that "Summarise this in three points" and "Give me a three-point summary" may produce systematically different outputs. This fragility is a fundamental property of LLMs — they are not executing formal logic. Production prompts must be evaluated across paraphrases and edge cases, not just the single wording that worked in development.

Q62. What is an "evaluation prompt" (also called an "LLM-as-judge" prompt)?

A. A prompt used to evaluate the developer's prompting skills against a benchmark.

B. A system prompt configuration used exclusively in model evaluation benchmarks.

C. A prompt that retrieves evaluation metrics from an external scoring service.

D. A prompt that instructs a language model to score, rank, or critique another model's output against defined criteria.

Correct Answer: D

Explanation

LLM-as-judge is a scalable evaluation technique: instead of manual human review or rigid regex matching, you prompt a capable model to evaluate outputs. "Score this response on helpfulness (1–5), accuracy (1–5), and tone (1–5). Explain each score." This enables systematic evaluation of open-ended outputs at scale — though the judge model also introduces bias.

Q63. What is "prompt regression testing" and why is it necessary?

A. A technique for identifying prompts that cause the model to regress to earlier, less capable behaviour.

B. Running a fixed test suite of prompts and expected outputs after any change to the prompt, model, or system to detect quality degradations before they reach production.

C. A method for automatically rolling back prompt changes that reduce user engagement metrics.

D. Testing whether a new model version responds consistently to prompts designed for an older model.

Correct Answer: B

Explanation

Prompts in production interact with a complex system: the prompt itself, the model version, retrieved content, and user input all affect outputs. When any of these change — prompt edit, model upgrade, data update — regression testing ensures previously working behaviour is preserved. Without it, silent quality degradation goes undetected.

Q64. What is the "evaluation rubric" approach to assessing model outputs?

A. A rubric is an automated scoring algorithm that computes quality scores from output tokens.

B. A grid of benchmark tasks sourced from academic evaluation datasets.

C. Defining explicit, observable criteria for what constitutes a good output, allowing consistent scoring across evaluators and over time.

D. An evaluation technique that compares model outputs to a golden reference answer using BLEU score.

Correct Answer: C

Explanation

Rubrics make subjective quality judgements explicit and repeatable. Instead of "is this a good answer?", a rubric specifies: "Does it address all parts of the question? (1 point) Is it factually accurate? (1 point) Is it appropriately concise? (1 point)." This structure enables consistent human evaluation and can be operationalised as an LLM-as-judge prompt.

Q65. What is "adversarial prompting" in the context of reliability testing?

A. Deliberately crafting inputs designed to break, confuse, or manipulate the model — used to find failure modes before they appear in production.

B. A technique for generating adversarial training examples to improve model robustness.

C. A method for testing whether the model can detect and resist adversarial user inputs.

D. The practice of using aggressive language in prompts to force the model to be more direct.

Correct Answer: A

Explanation

Before deploying a prompt, stress-test it: try confusing inputs, contradictory instructions, off-topic questions, injection attempts, and edge cases. If the model fails on adversarial inputs in testing, it will fail in production. Adversarial testing is not optional for applications that handle untrusted user input.

Q66. What is the difference between "precision" and "recall" when evaluating LLM classification outputs?

A. Precision is the proportion of outputs that are factually accurate; recall is the proportion that are relevant.

B. Precision measures output conciseness; recall measures how much of the input the model references.

C. Precision measures how often the model's positive labels are correct; recall measures how often actual positives are correctly identified.

D. Precision and recall are interchangeable metrics — they differ only in which class is treated as positive.

Correct Answer: C

Explanation

Precision: of the cases the model labelled positive, what fraction truly are? Recall: of the truly positive cases, what fraction did the model catch? For high-stakes classification (fraud detection, safety filtering), the trade-off between precision and recall has real consequences. Improving one often degrades the other at a fixed decision threshold.

Q67. What is "hallucination rate" as an evaluation metric?

A. The rate at which the model generates content outside its defined persona or scope.

B. A measure of how often the model fails to respond to a prompt within the defined context window.

C. The frequency with which the model's outputs differ across multiple runs of the same prompt.

D. The proportion of model outputs that contain factually incorrect or fabricated claims — measured by comparing outputs against ground truth or retrieved sources.

Correct Answer: D

Explanation

Hallucination rate is a key reliability metric for factual applications. It requires a ground truth — verified facts, retrieved source documents, or expert review — to assess whether the model's claims are accurate. Lowering hallucination rate typically involves grounding (RAG), output constraints, and citation requirements.

Q68. What is "A/B testing" of prompts and what makes it valid?

A. A testing method where version A is the production prompt and version B is an evaluation-only prompt never shown to users.

B. Comparing two prompt variants against each other with real or representative traffic to measure which produces better outcomes on defined metrics.

C. Testing prompt A in development and prompt B in production to compare real-world and simulated performance.

D. A randomised test where users are shown two model responses and asked to choose the better one.

Correct Answer: B

Explanation

Valid prompt A/B testing requires: random assignment of conditions, sufficient sample size to detect meaningful differences, a pre-defined success metric, and controlled conditions (same model, same data, only the prompt varies). Without these, observed differences are noise. Like any experiment, it should be designed before running, not analysed post-hoc.

Q69. What is "reference-free evaluation" and when is it necessary?

A. Evaluating prompts without testing them against the target model, using a proxy model instead.

B. A method for evaluating factual accuracy without access to the original source documents.

C. Evaluating model outputs without a ground truth reference answer — necessary for open-ended tasks where no single correct answer exists.

D. An automated evaluation technique that does not require human annotators.

Correct Answer: C

Explanation

Many real-world tasks — "write a marketing email", "summarise this article" — have no single correct answer. Reference-free evaluation assesses quality on dimensions like coherence, relevance, tone, and helpfulness without comparing to a gold standard. LLM-as-judge, rubric-based scoring, and human preference ratings are all reference-free methods.

Q70. What is "prompt drift" and how can it be detected?

A. The gradual degradation in prompt effectiveness over time as the model version, data, or user behaviour changes — detected through ongoing evaluation against a fixed test suite.

B. The tendency for long prompts to lose coherence toward the end due to context window limitations.

C. A phenomenon where model outputs drift toward training data patterns when prompts are ambiguous.

D. The gradual divergence between a prompt's intended behaviour and what users actually ask.

Correct Answer: A

Explanation

Prompt drift is an operational risk: a prompt that works well today may underperform after a model update, a change in retrieved content, or shift in user query patterns. The only way to detect it is continuous evaluation — running the same test suite regularly and tracking metric trends. Without monitoring, drift goes undetected until users complain.

Q71. What is the purpose of a "golden dataset" in prompt evaluation?

A. A training dataset used to fine-tune the model on the specific task the prompt is designed for.

B. A curated set of representative inputs with high-quality human-verified expected outputs used as a benchmark for evaluating and comparing prompt versions.

C. A set of adversarial inputs specifically designed to break the model under evaluation.

D. The original training data used by the model provider, used as a reference for capability assessment.

Correct Answer: B

Explanation

A golden dataset is the evaluation foundation: inputs that represent real-world usage, with outputs verified by domain experts. When you change a prompt, run the golden dataset and compare. Regression against golden examples is the most reliable signal of whether a prompt change is an improvement or a regression.

Q72. What is "inter-rater reliability" and why does it matter for LLM evaluation?

A. The reliability of the model's outputs when the same prompt is sent to multiple server instances simultaneously.

B. A metric for how consistently the model follows instructions across different evaluators' test inputs.

C. The agreement rate between the model's outputs and a reference answer database.

D. The degree to which different human evaluators agree on quality scores — low agreement means the evaluation criteria are ambiguous, making scores unreliable as a benchmark.

Correct Answer: D

Explanation

If two evaluators score the same output a 4 and a 2, your evaluation is measuring evaluator variance, not output quality. High inter-rater reliability — achieved through clear rubrics, calibration sessions, and examples — is a prerequisite for meaningful human evaluation. Without it, your evaluation data is noise.

Q73. What is "output toxicity evaluation" in the context of prompt testing?

A. A technique for filtering toxic words from model outputs using a keyword blocklist.

B. An evaluation that measures how often the model refuses to answer questions it should answer.

C. Systematically testing whether prompts produce harmful, biased, or offensive outputs across a representative range of inputs including adversarial cases.

D. A safety audit of the model provider's training data for toxic content.

Correct Answer: C

Explanation

Toxicity evaluation tests whether your application — including your system prompt — reliably prevents harmful outputs under realistic conditions. This means testing with adversarial inputs, sensitive topic probes, and jailbreak attempts. It is not sufficient to test only on polite, cooperative user inputs. Real users will push boundaries.

Q74. What is "calibration" in the context of model confidence and evaluation?

A. Calibration refers to adjusting the model's temperature so its outputs match the desired level of certainty.

B. A well-calibrated model's stated confidence accurately reflects its actual accuracy — when it says 80% confident, it is right about 80% of the time.

C. A post-processing technique that normalises confidence scores across different model outputs.

D. Calibration is the process of fine-tuning a model to produce correct answers on a validation set.

Correct Answer: B

Explanation

Calibration is a statistical property: a perfectly calibrated model's confidence scores are empirically accurate. LLMs are generally poorly calibrated — they can be confidently wrong. This matters for evaluation: a model saying "I'm 95% certain" should not be taken at face value without empirical calibration data.

Q75. What is the most important principle for maintaining reliable prompt behaviour in production?

A. Locking the prompt and never changing it once it has been tested and deployed.

B. Using the largest available model to maximise output quality and minimise failure rates.

C. Ensuring the system prompt is as long and detailed as possible to cover all possible scenarios.

D. Continuous evaluation against a fixed test suite — monitoring output quality metrics over time and treating prompt performance as an ongoing operational concern, not a one-time setup.

Correct Answer: D

Explanation

Prompts exist in a dynamic environment: models update, data changes, user behaviour shifts, edge cases emerge. The only way to maintain reliability is to treat prompt evaluation as an ongoing process — test suites, monitoring, regression checks, and periodic human review. One-time setup is not enough for production-quality AI systems.

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Your turning point in the AI era.

END OF QUESTION BANK

Your turning

point starts

here.